Marwan Shafi

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Education

Indian Institute of Information Technology (IIIT), Sonepat — Sonepat, India
Bachelor of Technology in Computer Science — *Aug 2023 – Jun 2027*

The Model School Abu Dhabi — Abu Dhabi, UAE
Senior Secondary (XII) CBSE — *Apr 2021 – Mar 2023*

Progressive English School Wadihuda — Kerala, India
Secondary (X) CBSE — *Jun 2020 – Mar 2021*

Experience

Kiwisoft Solutions — Kannur, Kerala
Fullstack Developer Intern — *Jun 2026 – Jul 2026*

- Built two production-grade Spring Boot systems an Employee & Department Management System and a Recruitment & Job Portal spanning 10+ modules including Authentication, RBAC, Leave Management, Job Postings, and Audit Logging
- Designed 32+ secure RESTful APIs with Spring Security and JWT access/refresh tokens, enforcing role-based access control across multiple user roles, and architected normalized PostgreSQL schemas across 18+ entities via Spring Data JPA/Hibernate
- Implemented multi-step approval workflows, centralized exception handling, and structured audit logging across both systems, validated all endpoints using Postman for functional and integration testing, mentored a fellow intern through debugging sessions on shared modules, and presented technical walkthroughs of both systems to the project lead for review

Technical Skills

Languages: Java, JavaScript, Python, SQL
Backend: Spring Boot, Spring Security, Spring Data JPA/Hibernate, Node.js, Express.js, WebSockets
Frontend: React.js, Next.js, Tailwind CSS, jQuery, HTML, CSS
Databases: PostgreSQL, MySQL, MongoDB, Redis
Auth & APIs: JWT, RBAC, Swagger/OpenAPI, RESTful APIs
AI/ML: Machine Learning, Deep Learning, Neural Networks, PyTorch, YOLOv8, LLM Fundamentals, RAG Basics
Tools: Docker, AWS EC2, Git, GitHub, Postman
CS Fundamentals: DSA, OOP, DBMS, Computer Networks, Operating

Projects

NutriVerse | *TypeScript, Node.js, Express.js, React, MongoDB, Redis, LangGraph, Gemini API, AWS EC2* — *Aug 2026*

- Architected and deployed a full-stack AI-powered recipe and nutrition platform using TypeScript, Node.js/Express, and React, implementing manual Dependency Injection and HttpOnly-cookie JWT authentication, provisioned on AWS EC2 (Nginx, PM2, HTTPS) with Vercel frontend hosting and Upstash Redis for cross-origin sessions and caching
- Engineered an automated nutrition pipeline using the Gemini API for natural-language ingredient parsing, mapping outputs against the USDA FoodData Central REST API to compute 7 nutritional metrics per recipe, with Zod-validated schemas enforced end-to-end from parsing to database writes
- Developed a LangGraph-based agentic recipe assistant with conditional LLM tool-calling across a 4-node graph, where an explicit NO_MATCH signal from a MongoDB search tool lets the LLM autonomously decide to invoke a Gemini-backed fallback-generation tool, guarded by a recursion limit against infinite tool call loops

Recruitment & Job Portal | *Java, Spring Boot, React, PostgreSQL, Docker, WebSocket, JWT* — *Jul 2026*

- Delivered a multi-tenant Applicant Tracking System with a Spring Boot REST API 50+ endpoints across 8 modules and React 19 frontend, powering a 6-stage Kanban hiring pipeline across 11 relational PostgreSQL entities; packaged with Docker and hosted on Render, Vercel, and Neon PostgreSQL
- Secured stateless auth using dual JWT tokens with HttpOnly cookies and Bearer fallback, enforcing RBAC across 5 roles via Spring Security with CSRF protection
- Enabled real-time messaging via Spring WebSocket, STOMP, and SockJS with live chat re-ranking, presence tracking, and delivery receipts, plus duplicate-candidate prevention through standardized email/phone validation across indexed PostgreSQL records

Detection of Adversarial Attacks in AI-Generated Images | *Python, PyTorch, Flask, JavaScript, Matplotlib* — *Apr 2026*

- Trained a 3-layer CNN from scratch on the CIFAKE dataset (120,000 real and Stable Diffusion-generated images) to classify AI-generated images, achieving 94.53% test accuracy
- Implemented the Fast Gradient Sign Method (FGSM) to attack the classifier across 6 perturbation strengths, showing accuracy collapse from 94.53% to 6.14%, and analysed attack behaviour with ROC curves, confusion matrices and perturbation heatmaps
- Developed a pixel-variation analysis method to flag adversarially perturbed images using a calibrated threshold, and deployed the model through a Flask server with a Chrome extension for real-time classification of web images

Certifications & Achievements

- Solved 700+ problems on coding platforms including LeetCode (Max rating: 1624, top 15% globally) and CodeChef (rating: 1540+)
- Ranked 3rd in "Guardians of the Algorithms" Intra-College Codathon organized by IIIT Sonepat
- Certified Cyber Engineer - Computer & Network Security (CCE-CNS), IIT Madras Pravartak - Proficiency Level A+